

Toll Traffic Analytics & Anomaly Detection

Implementation Roadmap — Palanpur Toll Plaza | 8-Step Plan with Ownership

#	Step	What It Involves	Responsibility	Timeline
1	Gather 4 years of raw data	Pull the full transaction export (like the sample file) for all 3 shifts, every day, for all 4 years, from the toll software vendor. One file per day/shift is fine — no need to merge it upfront.		
2	Get the extra context files	Collect tariff rate history (what rates applied when), any lane changes, system downtime days, and the holiday/mela calendar. Hand all of this to whoever builds the model — it cannot work without it.		
3	Hand off to a data person to clean and merge	A technical resource combines everything into one consistent table, fixes mismatched vehicle-class names across years, and marks known outage days so they aren't mistaken for anomalies.		
4	Build a simple baseline first	Before any AI/ML, establish a basic "normal range" — average traffic and revenue for each hour and day-of-week, with upper/lower bounds. This alone flags obvious anomalies and takes days, not weeks.		
5	Review the baseline against known events	Check whether it correctly spikes during Diwali and the Ambaji mela, and stays flat on a normal Tuesday. If it misfires, the data cleaning step needs another pass.		
6	Layer on the forecasting model	Once the baseline works, a data scientist adds a forecasting model (predicts tomorrow's expected traffic/revenue) and a more sensitive anomaly detector on top to catch subtler patterns like class-mix drift.		
7	Set alert thresholds with your team	Decide how sensitive flags should be — too tight causes false alarms daily; too loose lets real leakage slip through. This is a business judgment call, not a technical one.		
8	Pilot on one plaza, then scale	Run it live on this one plaza for a month, compare its flags against what the audit/vigilance team already knows, then extend to other plazas once trust is established.		

Note: Responsibility and Timeline columns left blank for internal assignment.